

# Filters in Tableau

A **filter** in Tableau is used to **show only the data you want to see** and hide the rest.

For example, if you have sales data for different cities, you can apply a **City filter** and select only **Mumbai**. Tableau will then show data only for Mumbai.

## Show Filter

When you add a field to the **Filters** shelf, you can select **Show Filter** to display the filter on the worksheet/dashboard.

The user can then interact with the filter without changing the data source.

## Common Filter Display Types

| Filter type              | Explanation                                 | Example                                                                                                                       |
|--------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Single Value             | Select only one value                       | Select <b>Mumbai</b>                                                                                                          |
| Multiple Values          | Select more than one value                  | Select <b>Mumbai + Pune</b>                                                                                                   |
| Multiple Values Dropdown | Select multiple values from a dropdown list | <input checked="" type="checkbox"/> Mumbai, <input checked="" type="checkbox"/> Pune                                          |
| Single Value Dropdown    | Select one value from a dropdown            | Mumbai ▼                                                                                                                      |
| Slider                   | Select a value or range by moving a slider  | Sales: ₹10,000 — ₹50,000                                                                                                      |
| Checkbox / List          | Select values using checkboxes              | <input checked="" type="checkbox"/> Mumbai <input checked="" type="checkbox"/> Pune <input checked="" type="checkbox"/> Delhi |

## Simple example

Suppose your data contains:

| City   | Sales  |
|--------|--------|
| Mumbai | 50,000 |
| Pune   | 30,000 |
| Delhi  | 40,000 |

If you apply a **City filter** and select **Mumbai**, Tableau displays:

| City   | Sales  |
|--------|--------|
| Mumbai | 50,000 |

So, the basic idea is:

**Filter = Decide which data you want to see.**

## Calculated Fields

A **calculated field** is used to create a **new field/value using existing fields**.

Think of it like creating a new value from existing values.

For example:

**Sale A + Sale B = Sale C**

If:

- Sale A = ₹10,000
- Sale B = ₹20,000

Then the calculated field can create:

**Sale C = ₹30,000**

In Tableau, you create a formula using existing fields, and Tableau calculates the result for you.

## Simple example

If you have:

[Sale A] and [Sale B]

You can create:

[Sale A] + [Sale B]

This creates a **new calculated field** that gives the combined sales.

In simple terms:

- **Filter** → controls what data you see
- **Calculated Field** → creates a new value from existing data